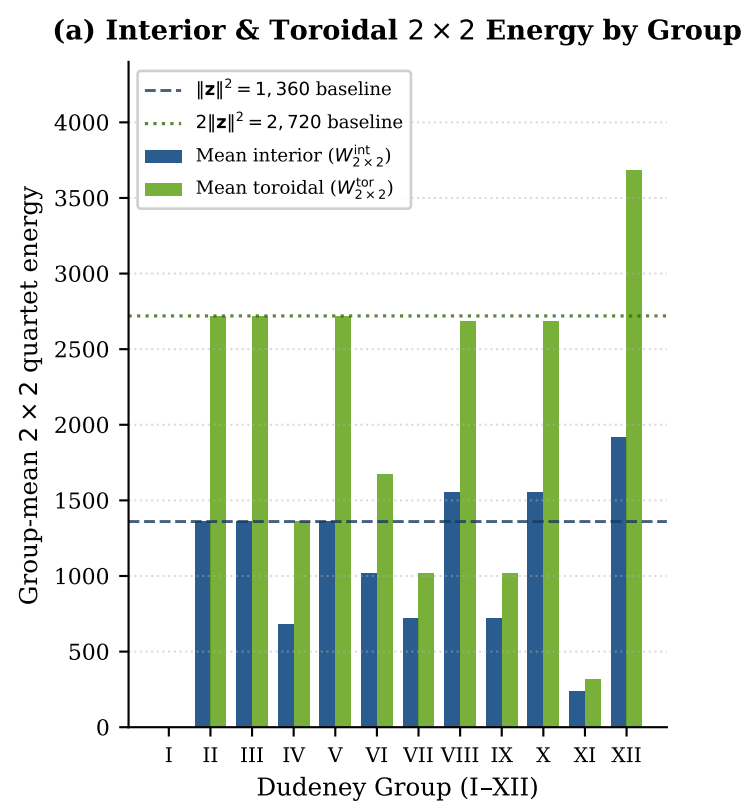
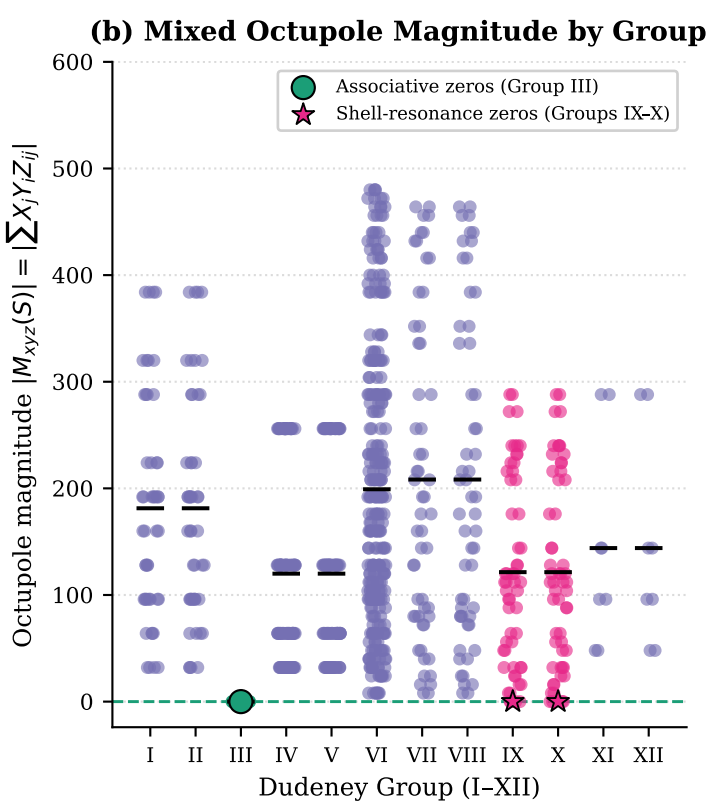
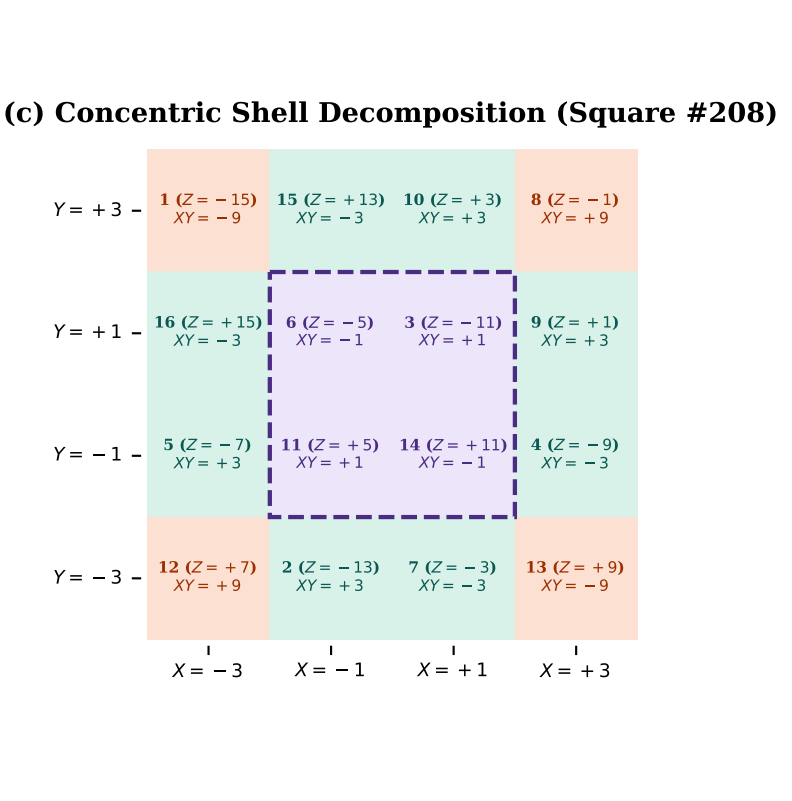

(a) Interior & Toroidal 2 × 2 Energy by Group
$\|\mathbf{z}\|^2 = 1,360$ baseline
$2\|\mathbf{z}\|^2 = 2,720$ baseline
Mean interior ($W^{\mathrm{int}}_{2\times2}$)
Mean toroidal ($W^{\mathrm{tor}}_{2\times2}$)
Group-mean 2 × 2 quartet energy
0
500
1000
1500
2000
2500
3000
3500
4000
I II III IV V VI VII VIII IX X XI XII
Dudeney Group (I–XII)


(b) Mixed Octupole Magnitude by Group
Associative zeros (Group III)
Shell-resonance zeros (Groups IX–X)
Octupole magnitude $|M_{xyz}(S)| = |\sum X_i Y_i Z_i|$
0
100
200
300
400
500
600
I II III IV V VI VII VIII IX X XI XII
Dudeney Group (I–XII)


(c) Concentric Shell Decomposition (Square #208)
$Y = +3$
$Y = +1$
$Y = -1$
$Y = -3$
1 ($Z = -15$) $XY = -9$
15 ($Z = +13$) $XY = -3$
10 ($Z = +3$) $XY = +3$
8 ($Z = -1$) $XY = +9$
16 ($Z = +15$) $XY = -3$
6 ($Z = -5$) $XY = -1$
3 ($Z = -11$) $XY = +1$
9 ($Z = +1$) $XY = +3$
5 ($Z = -7$) $XY = +3$
11 ($Z = +5$) $XY = +1$
14 ($Z = +11$) $XY = -1$
4 ($Z = -9$) $XY = -3$
12 ($Z = +7$) $XY = +9$
2 ($Z = -13$) $XY = +3$
7 ($Z = -3$) $XY = -3$
13 ($Z = +9$) $XY = -9$
$X = -3$
$X = -1$
$X = +1$
$X = +3$